Highlights

Same weight, different loss: the weak-form discretization sets the effective initial-condition weight in physics-informed training

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- Conflicting boundary and initial data push both losses away from the solution
- The weak loss weights the initial condition 10^3 to 10^5 times more strongly
- On fine meshes the ratio is test functions per element times E^{2n}
- Weight sweeps and mesh changes move each loss's failure as the ratio predicts
- A warm-start test exposes the failure while the training loss falls

Same weight, different loss: the weak-form discretization sets the effective initial-condition weight in physics-informed training

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ABSTRACT

Physics-informed solvers for time-dependent problems often build the Dirichlet data into the trial space and impose the initial condition through a penalty. This paper gives a scaling law for the weight of that penalty in variational (weak-in-space) losses and a verification test that exposes its effect, with incompatible initial and boundary data as the stress case. When the data clash at $t = 0$, no smooth trial function meets both, and training on the physics loss moves a supervised fit to the exact solution away from it: a strong-form loss fails early in the transient and a weak-form loss late. Where the model fails is set by the effective weight of the penalty. The same nominal weight acts 10^3 to 10^5 times more strongly in the weak loss; we call this factor κ. A scaling lemma shows that on fine meshes κ equals the number of test functions per element times E^{2n}, with E elements per dimension in n dimensions, and its error tracks the measured variation of the residual across an element. In 1D, weight sweeps and changes of mesh and test-function order move each loss's failure as κ predicts, on heat and advection–diffusion problems with exact solutions; in 2D the sweeps meet within a factor of about two of κ. Normalizing the weak residual by element measure removes the mesh dependence. Loss weights therefore do not transfer between loss forms or between meshes.

1. Introduction

Physics-informed neural networks place the governing equation in the loss function [1]. Variational versions project the residual onto a finite set of test functions and so work with a weak form of the equation [2–4]. In both, Dirichlet data can be imposed exactly by building the trial space around a distance function [5, 6]. Exact imposition was the most accurate of four Dirichlet strategies compared for PINNs and VPINNs [7]. For a time-dependent problem, the initial condition is then usually left to the loss as a penalty. This split is close to a default. Such models are used where a classical solver is not a direct substitute, for example in inverse problems, parametric surrogates and digital twins that combine measured data with physics.

Sometimes the data disagree. Consider a plate at temperature 0 whose edge is suddenly held at temperature 1. At the edge, at $t = 0$, the boundary requires 1 and the initial condition requires 0. Such incompatible data are a classical topic for parabolic equations. The solution is not smooth at the corner, and higher regularity needs compatibility conditions [8]. Classical discretizations have known treatments. Error estimates for Galerkin methods with nonsmooth initial data degrade as $t \to 0$ [9], and Crank–Nicolson time stepping needs a few damping steps at the start to recover its order [10, 11]. A neural trial space with an exact boundary has a different problem. It is smooth and pinned to the boundary value, so it cannot represent the jump. The penalty can be driven toward zero only through an ever thinner layer at the edge, which a network of fixed size cannot follow. Over the models the network can reach, the penalty and the physics residual cannot both be made small, and the penalty keeps pulling.

Competing loss terms are a known source of failure in physics-informed training [12–14], and reweighting is the standard response. These studies balance terms that can all be satisfied at once. Causal training [15] reweights the residual in time so that early times are fitted before later ones; it acts on the same time axis on which the present failure appears. Corner conflicts are known to hurt training, and the usual remedy is to drop points near the discontinuity from the loss [16]. For weak losses, Rojas et al. [17] showed that the plain variational loss need not track the error and proposed a dual-norm loss that does for a rich enough test space. Robust variational losses for time-dependent problems impose the initial condition through the trial space, which assumes compatible data [18]. A time-discrete

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variational scheme was applied to a freezing problem whose boundary and initial temperatures differ [19], a case of the same clash in practice.

This paper gives two tools for this setting: a verification test and a scaling law for loss weights in variational discretizations. The test is a code-verification test in the sense of [20, 21]: a known solution is supplied and the method is checked against it. It fits a network to the exact solution by supervision and then trains it on the physics loss alone. Some drift is expected whenever the exact solution lies outside the trial space; the test measures how large it is and where in the transient it falls. Under the clash both loss forms drift, at opposite ends of the transient, while the training loss falls. Where the failure falls is set by the effective weight of the initial-condition penalty. The weak-form discretization multiplies this weight by a factor κ, which a scaling lemma predicts on fine meshes and which weight sweeps, changes of mesh and test-function order, and a second equation confirm. The contributions are as follows.

- A warm-start verification test that measures how far, and where in time, training leaves a supervised fit (Section 5.1).
- A scaling law for the ratio κ of the effective weights in the two losses, $\kappa = M E^{2n}$ on fine meshes, with a measured criterion for when a mesh is fine enough (Section 3.4).
- Interventions on the weight, the mesh, the test-function order and the residual normalization, on two equations, all of which move the failure as κ predicts (Sections 5.2–5.4).
- A classical baseline that places the problem in the trial space, and a comparison of remedies (Sections 5.5 and 6).

Section 2 states the problems, the trial space and the losses. Section 3 defines the effective weight and gives the proposition and the scaling lemma. Section 4 describes the experiments, Section 5 reports the results and Section 6 the remedies. Section 7 discusses the results, and Section 8 concludes.

2. Problem formulation and discretization

2.1. Model problems

The heat equation is solved on $\Omega = (0,1)^n$, $n \in \{1,2\}$, for $t \in (0,1]$:

$$\partial_t u = a\,\Delta u \text{ in } \Omega, \qquad u = 1 \text{ on } \partial\Omega, \qquad u(\cdot,0) = u_0, \tag{1}$$

with $a = 0.1$. Two initial conditions are used. The *matching* one, $u_0 = 1 - \prod_i \sin(\pi x_i)$, equals 1 on $\partial\Omega$, and the exact solution is $u = 1 - e^{-na\pi^2 t} \prod_i \sin(\pi x_i)$. The *clashing* one is $u_0 = 0$. Its exact solution is a Fourier sine series with 400 odd modes per dimension, accurate for $t \geq 10^{-3}$.

The second problem is the advection–diffusion equation

$$\partial_t u + c\,\partial_x u = a\,\partial_{xx} u \text{ on } (0,1), \qquad u(0,t) = u(1,t) = 1, \tag{2}$$

with $c = 1$, $a = 0.1$ and the same two initial conditions. The substitution $u = 1 - e^{cx/(2a) - c^2 t/(4a)}\, w$ turns Eq. (2) into $w_t = a\, w_{xx}$ with $w = 0$ at both ends and $w(x,0) = e^{-cx/(2a)}\big(1 - u_0(x)\big)$, which is solved with a sine series of 799 modes. This reference has a PDE residual of about 10^{-14} and agrees with an independent Crank–Nicolson solve to about 10^{-6}.

2.2. Trial space with an exact boundary

The trial functions are

$$u_\theta(x,t) = 1 + B(x)\, N_\theta(x,t), \qquad B(x) = \prod_{i=1}^{n} 4x_i(1 - x_i), \tag{3}$$

where N_θ is a fully connected network with parameters θ: input $n+1$, five hidden layers of width 64, SiLU activations, double precision. Every u_θ equals 1 on $\partial\Omega$ for every t.

Definition 1 (Clash). The initial and boundary data *clash* if $u_0 \neq 1$ on $\partial\Omega$. Since $u_\theta(\cdot,0)$ is continuous and equals 1 on $\partial\Omega$, no trial function then meets the initial condition, and the continuous penalty $\int_\Omega \big(u_\theta(x,0) - u_0(x)\big)^2 dx$ is positive for every θ.

Because S is redrawn at every step, the expected value of L_{IC} is this continuous penalty; a network cannot drive it to zero by fitting a fixed set of points. The infimum of the loss over the closure of the trial space is nevertheless zero. The time-shifted solution $u(x, t+\varepsilon)$ equals 1 on $\partial\Omega$, satisfies Eq. (1) exactly, and has an initial-condition error that vanishes as $\varepsilon \to 0$. The infimum is not attained: the shifted solution has a boundary layer of width of order $\sqrt{a\varepsilon}$, thinner the smaller the error. A network of fixed size trained for a fixed number of steps cannot follow that layer, so over the models it reaches, the two terms cannot both be made small. Proposition 1(b) describes the resulting trade-off for a fixed trial space.

2.3. Losses

The loss is

$$L(\theta) = \lambda_{\mathrm{phys}}\, L_{\mathrm{phys}}(\theta) + \lambda_{\mathrm{IC}}\, L_{\mathrm{IC}}(\theta), \qquad L_{\mathrm{IC}}(\theta) = \frac{1}{|S|} \sum_{z \in S} \big(u_\theta(z,0) - u_0(z)\big)^2, \tag{4}$$

with S a set of 500 points drawn uniformly in Ω and redrawn at every step, $\lambda_{\mathrm{phys}} = 12$ and $\lambda_{\mathrm{IC}} = 3$. Two physics terms are compared.

Strong form The pointwise residual is averaged over a set Q of space–time points:

$$L_{\mathrm{s}}(\theta) = \frac{1}{|Q|} \sum_{(x,t) \in Q} \big(\partial_t u_\theta - a\, \Delta u_\theta\big)^2, \tag{5}$$

with $|Q| = 1{,}280$ in 1D and 16,000 in 2D, drawn uniformly in $\Omega \times (0,1)$ and redrawn at every step.

Weak form The domain is split into uniform elements K_e, with $E = 16$ elements in 1D and 8×8 in 2D. On each element, test functions $v_{e,i}$ are built from $P_{m+1} - P_{m-1}$, with P_m the Legendre polynomials mapped to K_e, and from tensor products of these in 2D. The order p means $m = 1, \ldots, p+1$, so there are $M = (p+1)^n$ test functions per element and the highest polynomial degree is $p+2$. The defaults are $p = 6$ in 1D ($M = 7$) and $p = 4$ in 2D ($M = 25$). Each test function vanishes on ∂K_e. Multiplying Eq. (1) by $v_{e,i}$ and applying Green's first identity on K_e gives the residual

$$R_{e,i}(t_j) = \int_{K_e} \big(\partial_t u_\theta\, v_{e,i} + a\, \nabla u_\theta \cdot \nabla v_{e,i}\big)\, dx, \tag{6}$$

evaluated by Gauss–Legendre quadrature (8 points per element in 1D, 5×5 in 2D, raised to $p+2$ per dimension for higher orders) at 10 times t_j drawn uniformly in $(0,1)$ at every step. The weak residual needs only first derivatives of u_θ; the strong residual needs second derivatives. On the default meshes the two losses evaluate the network at the same number of points per step. The weak loss is the mean square,

$$L_{\mathrm{w}}(\theta) = \frac{1}{N_R} \sum_{e,i,j} R_{e,i}(t_j)^2, \tag{7}$$

with N_R the number of terms. Each time is squared separately; there is no sum over time before squaring. The loss is therefore weak in space and collocated in time: $\partial_t u_\theta$ enters pointwise at the sampled times, and there are no test functions in time. For advection–diffusion, $c\, \partial_x u_\theta\, v_{e,i}$ is added to the integrand of Eq. (6) and $c\, \partial_x u_\theta$ to the strong residual.

Two variants of the weak loss are also used. The *dual-norm* loss [17] replaces the mean of squares at each time by $R(t_j)^\top G^{-1} R(t_j)$, with G the H^1-seminorm Gram matrix of the same test functions. Its condition number is about 4 in 1D and 7 in 2D. The *normalized* loss divides each $R_{e,i}$ by the element measure $|K_e|$ before squaring. With E uniform elements in 1D, this multiplies L_{w} by E^2.

Neither loss uses gradient clipping. As in FastVPINNs [4, Eq. 9], the cell-wise residual of Eq. (6) is an integral over the element and is not divided by the element measure. Codes differ in how they combine the cell-wise residuals. Eq. (10) of [4] averages the squared residual vector over elements, while its Algorithms 2 and 3, and the hp-VPINN implementation it describes (its Algorithm 1), average over the test functions of each element and sum over elements. Eq. (7) averages over both.

2.4. Error measure

The error is the relative error of the transient,

$$e(t) = \frac{\|u_\theta(\cdot,t) - u(\cdot,t)\|_2}{\|u(\cdot,t) - 1\|_2}, \tag{8}$$

with discrete L^2 norms on 201 points in 1D and 41×41 points in 2D, at $t \in \{0.01, 0.05, 0.1, 0.3, 1\}$. A model that outputs the steady state $u = 1$ scores 1. The denominator shrinks as the transient decays, which makes late errors look large. Forms are therefore compared only at the same time. Absolute errors, $e(t)$ times the RMS of $u - 1$, are given in Table 4.

3. The effective weight

3.1. Definition

Let $g_{\mathrm{IC}} = \nabla_\theta\big(\lambda_{\mathrm{IC}} L_{\mathrm{IC}}\big)$ and $g_{\mathrm{phys}} = \nabla_\theta\big(\lambda_{\mathrm{phys}} L_{\mathrm{phys}}\big)$, each averaged over $D = 100$ fixed draws of the sampled points. The 100-draw estimate is within 4% of a 1,000-draw one (supplementary material). The *effective weight* of the initial-condition term is

$$\eta = \frac{\|g_{\mathrm{IC}}\|}{\|g_{\mathrm{phys}}\|}, \tag{9}$$

measured at the supervised fit of Section 4.1. The initial-condition gradient is the same in both losses, since it is the same term on the same points. The physics gradient is not. The ratio between the forms is

$$\kappa = \frac{\|g_{\mathrm{phys}}\|_{\mathrm{strong}}}{\|g_{\mathrm{phys}}\|_{\mathrm{weak}}} = \frac{\eta_{\mathrm{weak}}}{\eta_{\mathrm{strong}}}. \tag{10}$$

Since η is linear in λ_{IC}, the weak loss at a given initial-condition weight has the same effective weight as the strong loss at κ times that weight. Adam rescales the gradient parameter by parameter, so a ratio of global norms does not fix the update on its own. When the weak physics gradient is the strong one divided by κ, the total weak-loss gradient is $1/\kappa$ times the strong-loss gradient with the initial-condition weight multiplied by κ. Adam is invariant to such an overall rescaling when the gradient is large compared with its stabilizing constant [22], so it then takes the same steps for both. The two physics gradients are close to parallel at the supervised fits (cosine at least 0.987, Section 3.4), and the sweeps of Section 5.2 test the resulting prediction directly.

3.2. A linear trial space

For a trial space that is linear in its parameters, the role of κ can be stated exactly. Let $u_\theta = 1 + B(x)\,\Phi(x,t)\,\theta$ with a fixed feature map Φ. Every residual is then affine in θ. Collect the physics residuals (pointwise values for the strong form, the $R_{e,i}(t_j)$ for the weak form) as $A\theta - b$ and the initial-condition residuals as $C\theta - d$, with the averaging factors absorbed into A and C. The loss of Eq. (4) becomes

$$L(\theta) = \lambda_{\mathrm{phys}}\,\|A\theta - b\|^2 + \lambda_{\mathrm{IC}}\,\|C\theta - d\|^2. \tag{11}$$

Assume $\ker A \cap \ker C = \{0\}$. In this setting the data clash when no θ satisfies both $A\theta = b$ and $C\theta = d$.

Proposition 1. *Let $\rho = \lambda_{\mathrm{IC}}/\lambda_{\mathrm{phys}} > 0$.*

(a) L has the unique minimizer

$$\theta^*(\rho) = \big(A^\top A + \rho\, C^\top C\big)^{-1}\big(A^\top b + \rho\, C^\top d\big), \tag{12}$$

which depends on the weights only through ρ.

(b) The physics term $P(\rho) = \|A\theta^*(\rho) - b\|^2$ is nondecreasing in ρ, and the initial-condition term $I(\rho) = \|C\theta^*(\rho) - d\|^2$ is nonincreasing. Under a clash, $P(\rho)$ and $I(\rho)$ are never both zero.

(c) If the weak and strong residuals are proportional in the mean-square sense, $A_{\mathrm{w}}^\top A_{\mathrm{w}} = s^2 A_{\mathrm{s}}^\top A_{\mathrm{s}}$ and $A_{\mathrm{w}}^\top b_{\mathrm{w}} = s^2 A_{\mathrm{s}}^\top b_{\mathrm{s}}$ for a scalar $s > 0$, then $\kappa = 1/s^2$ at every θ, and $\theta_{\mathrm{w}}^*(\rho) = \theta_{\mathrm{s}}^*(\kappa\rho)$.

PROOF. (a) Dividing L by λ_{phys} does not move its minimizer, so only ρ matters. The Hessian of $\|A\theta-b\|^2+\rho\|C\theta-d\|^2$ is $2(A^\top A+\rho\, C^\top C)$, which is positive definite because $\ker A\cap\ker C=\{0\}$. Setting the gradient to zero gives Eq. (12).

(b) Take $\rho_1<\rho_2$ with minimizers θ_1,θ_2, and write $P_k=P(\rho_k)$, $I_k=I(\rho_k)$. Optimality of each minimizer gives $P_1+\rho_1 I_1\le P_2+\rho_1 I_2$ and $P_2+\rho_2 I_2\le P_1+\rho_2 I_1$. Adding the two inequalities gives $(\rho_2-\rho_1)(I_2-I_1)\le 0$, so $I_2\le I_1$. The first inequality then gives $P_1-P_2\le\rho_1(I_2-I_1)\le 0$, so $P_1\le P_2$. If $P(\rho)=I(\rho)=0$ for some ρ, then $\theta^*(\rho)$ satisfies $A\theta=b$ and $C\theta=d$, which a clash excludes.

(c) The two residual vectors may have different lengths; only their normal equations enter. The physics gradient of the weak loss is $2\lambda_{\text{phys}}(A_{\text{w}}^\top A_{\text{w}}\theta-A_{\text{w}}^\top b_{\text{w}})=s^2\cdot 2\lambda_{\text{phys}}(A_{\text{s}}^\top A_{\text{s}}\theta-A_{\text{s}}^\top b_{\text{s}})$, and the initial-condition gradient is the same in both losses, so $\kappa=1/s^2$. Substituting into Eq. (12) gives $\theta^*_{\text{w}}(\rho)=(s^2A_{\text{s}}^\top A_{\text{s}}+\rho\, C^\top C)^{-1}(s^2A_{\text{s}}^\top b_{\text{s}}+\rho\, C^\top d)=\theta^*_{\text{s}}(\rho/s^2)=\theta^*_{\text{s}}(\kappa\rho)$.

Parts (a) and (b) are standard results for weighted least squares and scalarized two-objective problems, the property behind the L-curve of Tikhonov regularization [23, 24]; part (c) is a rescaling identity. Part (b) uses only optimality, so it holds for the global minimizers of any loss with two weighted terms: raising the weight trades physics error for initial-condition error and never improves both. Part (c) is the comparison tested in Section 5.2: the weak loss behaves like the strong loss with its initial-condition weight multiplied by κ.

Remark 1. The proportionality assumed in part (c), and the size of κ, follow from Lemma 1 for residuals that are smooth on the element scale. The step from loss terms to the error early and late in the transient is made by the experiments of Section 5, for the network trial space.

3.3. Check on a linear model

The proposition was checked on the clashing 1D heat problem with a linear trial space of 140 Legendre features in (x,t) and the exact boundary of Eq. (3). The closed form and part (c) reproduce a direct least-squares solve to rounding error, and for the strong, weak and dual-norm losses P rises and I falls monotonically with ρ. For the actual weak residual the linear model has $\kappa_{\text{lin}}\approx 1{,}090$, and the weight that minimizes its error differs between the two losses by a factor of 100 to 1,000, the same order. Polynomial features represent the corner layer well, so in this model the two terms barely conflict and the error stays small. The drift is a property of the network trial space and is studied with it in Section 5.

3.4. The size of κ

The size of κ follows from the structure of the weak loss when the residual is smooth on the element scale.

Lemma 1. *Let the weak loss of Eq. (7) use E uniform elements of width $h=1/E$ per dimension in n dimensions, with the $M=(p+1)^n$ test functions per element of Section 2.3, and let the element integrals be exact. Let $r=\partial_t u_\theta-a\Delta u_\theta$, let $\bar r_e$ be its value at the centre of K_e, and let ℓ be a length such that on every element and at every sampled time $|\partial_x^\alpha r|\le C\,|\bar r_e|\,\ell^{-|\alpha|}$ for $|\alpha|\le 2$, with the same bound for each component of $\nabla_\theta r$ relative to $\nabla_\theta\bar r_e$. Assume that both losses sample the same distribution of times, and let $\bar L$ be the space–time mean of r^2, which the strong loss of Eq. (5) estimates. Then*

$$L_{\text{w}}=\frac{h^{2n}}{M}\,\bar L\left(1+O\big((h/\ell)^2\big)\right),\qquad \nabla_\theta L_{\text{w}}=\frac{h^{2n}}{M}\Big(\nabla_\theta\bar L+\varepsilon\Big),\quad \|\varepsilon\|\le C'\,(h/\ell)^2\,G, \tag{13}$$

where G is the mean over elements and times of $|\bar r_e|\,\|\nabla_\theta\bar r_e\|$. If $\|\nabla_\theta\bar L\|$ is not small compared with G, then, up to the sampling error of the strong loss, $\kappa=M\,E^{2n}\left(1+O((h/\ell)^2)\right)$.

PROOF. Since every $v_{e,i}$ vanishes on ∂K_e, Green's first identity turns Eq. (6) into $R_{e,i}=\int_{K_e} r\,v_{e,i}\,dx$. Take $n=1$ first. Map K_e to $[-1,1]$ by $x=x_e+h\xi/2$, so that $v_{e,i}(x)=\varphi_m(\xi)$ with $\varphi_m=P_{m+1}-P_{m-1}$, and expand $r=\bar r_e+\frac{h}{2}r'_e\,\xi+\rho_e(\xi)$ about the element centre, where $|\frac{h}{2}r'_e|\le C|\bar r_e|(h/\ell)$ and $|\rho_e|\le C|\bar r_e|(h/\ell)^2$. Since $\int_{-1}^{1}P_k\,d\xi=2\delta_{k0}$, only $\varphi_1=P_2-P_0$ has a nonzero mean, $\int\varphi_1\,d\xi=-2$. By parity $\int\xi\varphi_1\,d\xi=0$, so $R_{e,1}=-h\,\bar r_e\big(1+O((h/\ell)^2)\big)$. For $m\ge 2$ the constant term integrates to zero and $|R_{e,m}|\le Ch|\bar r_e|(h/\ell)$. Averaging the squares over the M test functions, the elements and the sampled times gives $L_{\text{w}}=(h^2/M)\,\overline{\bar r_e^{\,2}}\,(1+O((h/\ell)^2))$, and the mean of r^2 at the element centres equals $\bar L$ to the same relative order by the midpoint rule. For the gradient, $\nabla_\theta L_{\text{w}}$ is the mean of $2R_{e,i}\nabla_\theta R_{e,i}$ with

Table 1
Checks of Lemma 1 at the supervised fits, warm start, seeds 42–44 (ranges over seeds; $(h/\ell)^2$: mean over seeds of the c_0^2-weighted mean). Cosines are between the averaged gradients over $D = 100$ draws; Γ is the cancellation ratio of the text. Ratio: measured κ (100 draws, GPU) over ME^{2n}; the last column is omitted where the deviation is within the noise. Default mesh: 16 elements, $p = 6$ in 1D; 8×8, $p = 4$ in 2D.

Setting	cos(weak, strong)	cos(IC, weak)	Γ	$(h/\ell)^2$	ratio	$(\text{ratio} - 1)/(h/\ell)^2$
1D clash, $E = 4$	0.987–0.994	−0.75 to −0.40	0.96–0.99	106	4.4	0.032
1D clash, $E = 8$	0.997–0.999	−0.77 to −0.46	0.96–0.99	40	2.9	0.047
1D clash, $E = 16$	0.9991–0.9996	−0.79 to −0.47	0.96–0.99	6.5	1.22	0.034
1D clash, $E = 32$	0.9998–0.9999	−0.80 to −0.47	0.96–0.99	1.3	0.99	–
1D matching, $E = 16$	1.000	+0.07 to +0.40	0.60–0.65	0.38	1.02	–
2D clash, 8×8	0.999–1.000	−0.96 to +0.63	0.84–0.86	17	2.1	0.067
2D matching, 8×8	0.999–1.000	+0.46 to +0.71	0.41–0.55	1.7	1.06	–

$\nabla_\theta R_{e,i} = \int_{K_e} \nabla_\theta r\, v_{e,i}\, dx$. The same expansion gives $2R_{e,1}\nabla_\theta R_{e,1} = 2h^2\, \bar{r}_e \nabla_\theta \bar{r}_e + O\big(h^2 (h/\ell)^2 |\bar{r}_e|\, \|\nabla_\theta \bar{r}_e\|\big)$ and bounds the terms with $m \geq 2$ by the same quantity. These errors are bounded term by term, not relative to the sum, which is why the gradient statement is additive: the sum $\nabla_\theta \bar{L}$ can be small through cancellation while each term is not. In two dimensions the test functions are products $\varphi_m(\xi)\varphi_{m'}(\eta)$, only $(m, m') = (1, 1)$ has a nonzero mean, and the Jacobian $h^2/4$ gives $R_{e,(1,1)} = h^2\, \bar{r}_e\, (1 + O((h/\ell)^2))$, hence the factor h^4/M. When $\|\nabla_\theta \bar{L}\|$ is not small compared with G, the ratio of gradient norms is $Mh^{-2n}(1 + O((h/\ell)^2)) = ME^{2n}(1 + O((h/\ell)^2))$.

On a fine mesh the weak residual is, to leading order, the element-centre residual scaled by h^n and carried by one test function per element. This is the proportionality assumed in Proposition 1(c), with $s^2 = h^{2n}/M$. The factor h^{2n} comes from integrating over small elements without dividing by their measure, and $1/M$ from averaging over M test functions, of which a smooth residual excites one. Both are fixed by the discretization. The same calculation gives κ for the other conventions of Section 2.3: $\kappa = E^{2n}$ when the squared residuals are averaged over elements and summed over test functions (Eq. 10 of [4]), and $\kappa = ME^n$ when they are averaged over test functions and summed over elements (Algorithms 2 and 3 of [4], and the hp-VPINN implementation they describe). Two codes with the same mesh and test space therefore differ in κ by a factor of M or E^n, and a weight tuned in one does not carry to the other.

Table 2 and Figure 1 compare the prediction with κ measured at 100 draws. With the mesh fixed at 16 elements, κ is 1.21 to 1.25 times $256(p + 1)$ for every order, so the order dependence is the predicted one; the linear model of Section 3.3 shows the same slope. On 16 and 32 elements the prediction is within 25% for both initial conditions, both equations and warm and cold starts. On coarser meshes κ is larger than ME^{2n}, by 4.4 on 4 elements, 2.9 on 8 and 2.1 on the 8×8 mesh in 2D, and the slope fitted over 4 to 32 elements is 1.24, rising to 1.69 between 16 and 32. The lemma is an asymptotic statement and needs h small compared with ℓ, the length over which the residual changes. On the default meshes $(h/\ell)^2$ is 6.5 in 1D and 17 in 2D (Table 1), so agreement there is measured rather than implied by the lemma; the relation below quantifies it.

Table 1 tests the lemma's hypotheses at the supervised fits. The size of h/ℓ is estimated on every element and sampled time from a Legendre fit of r at the element's quadrature points, $r \approx c_0 + c_1 P_1(\xi) + c_2 P_2(\xi)$, as $h/\ell_e = \max\big(2|c_1|/|c_0|, \sqrt{12|c_2|/|c_0|}\big)$ (the tensor version in 2D), and averaged with weights c_0^2. The weak and strong physics gradients point the same way, with cosine 0.987 or more on every mesh and 0.9999 on 32 elements, and the gradient shows no strong cancellation (Γ of 0.84 to 0.99 under the clash). The weighted $(h/\ell)^2$ falls with refinement, from about 106 on 4 elements to 1.3 on 32, and the deviation of κ from ME^{2n} tracks it: where the deviation is above the noise of a few percent, $\kappa/(ME^{2n}) - 1$ is 0.03 to 0.07 times $(h/\ell)^2$, on every 1D mesh and in 2D. The constant depends on the estimator; the relation does not. It gives a working criterion: $(h/\ell)^2$ below about 4 at a supervised fit or a reference solution keeps the prediction within roughly 25%, and estimating it needs only residual evaluations. The same relation accounts for the effect of the clash on κ. The clashing fits have the larger $(h/\ell)^2$ (6.5 against 0.4 in 1D, 17 against 1.7 in 2D) and the larger κ, by 20% in 1D and a factor of 2 in 2D.

The cosine between the initial-condition and physics gradients measures the clash directly. With matching data it is positive (+0.07 to +0.71): both terms pull the same way. Under the clash it is negative (−0.47 to −0.96), except for one 2D seed (+0.63): the initial-condition term pulls against the physics.

Table 2
Measured κ (100 draws, geometric mean over seeds 42–44, warm start unless stated) against Lemma 1, $\kappa \approx M E^{2n}$ with $M = (p+1)^n$. Default: 1D heat, clashing initial condition, 16 elements, $p = 6$ ($M = 7$, prediction 1,792). $\kappa_{\text{norm}} = \kappa / E^2$ is the ratio after normalizing the weak residual by element measure, for which the lemma predicts $M = 7$.

Elements E ($p = 6$)	4	8	16	32
κ	490	1,300	2,200	7,100
$7E^2$	112	448	1,792	7,168
ratio	4.4	2.9	1.23	0.99
κ_{norm} (lemma: 7)	30	20	8.6	6.9
Order p ($E = 16$)	2	4	6	10
κ	940	1,600	2,200	3,400
$256(p+1)$	768	1,280	1,792	2,816
ratio	1.22	1.25	1.23	1.21
Other settings		κ	prediction	ratio
1D heat, matching IC		1,810–1,820	1,792	1.01–1.02
1D heat, cold start		1,750–1,790	1,792	0.98–1.00
1D advection–diffusion		2,000	1,792	1.12
2D heat ($E = 8$, $p = 4$, $M = 25$)		2.1–2.3×10^5	1.02×10^5	2.0–2.2
2D heat, matching IC		1.06–1.11×10^5	1.02×10^5	1.04–1.08
2D heat, cold start		1.01–1.04×10^5	1.02×10^5	0.99–1.02

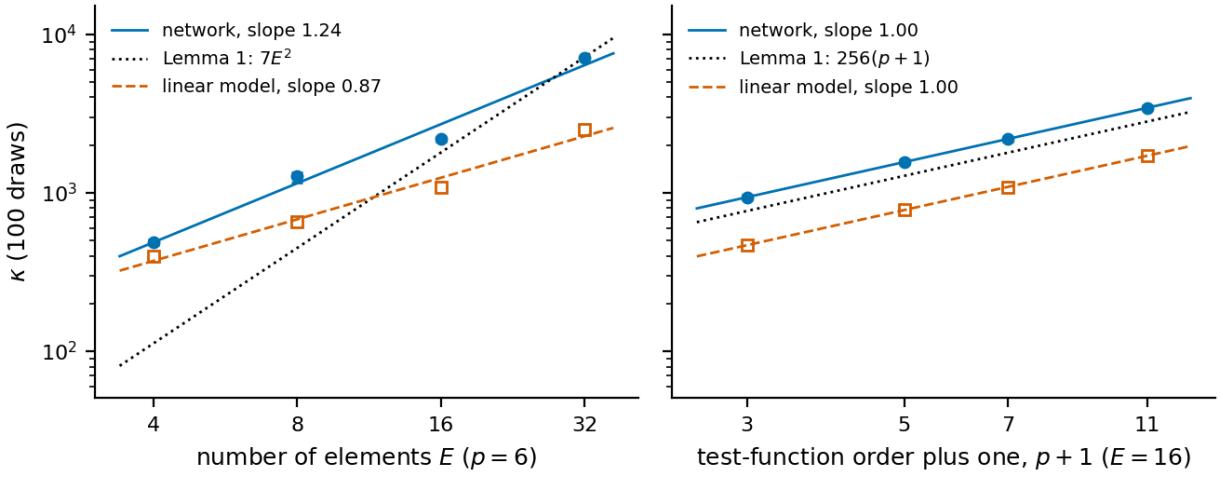


Figure 1: The weak-form discretization sets κ. Left: κ against the number of elements E ($p = 6$). Right: κ against the test-function order plus one ($E = 16$). 1D heat, clashing initial condition, warm start, 100 draws. Filled: network (geometric mean over seeds 42–44, bars: min–max). Open: linear model of Section 3.3. Solid and dashed lines: least-squares fits. Dotted: Lemma 1, $7E^2$ (left) and $256(p+1)$ (right).

4. Experimental design

4.1. Warm and cold starts

A *warm* start is a network fitted to the exact solution by supervision for 5,000 Adam steps, then trained on the loss of Eq. (4) for 10,000 steps at learning rate 10^{-4}. If the loss is minimized at the solution, training should not make the fit worse. With a matching initial condition it can improve it. The supervised fits are within about 0.07 of the exact solution at every time (Table 3). They are not exact, since the exact solution cannot be represented under the clash. A *cold* start is a random initialization trained for 15,000 steps at learning rate 10^{-3}. All runs use cosine decay.

Arms use seeds 42–44. The three 2D weak-form arms at default weights with the clash (plain and dual-norm from warm starts, plain from cold starts) use seeds 42–51, because their late-time spread is large (supplementary material).

4.2. Weight sweeps and the match criterion

The initial-condition weight is multiplied by a fixed factor in one form and compared with the other form at its default weight. The strong form is swept upward and the weak form downward, from the same warm starts, with seeds 42–44 and everything else unchanged. The multipliers were fixed in advance and are listed in the supplementary material. Sweep comparisons use the three-seed means of the default arms as reference, so like is compared with like.

The *gap* between two arms is the largest difference of their mean errors over the five times. Two arms *match* when the gap is at most 0.05. This threshold is above the seed range of every 1D weak-form arm at $t = 1$ (at most 0.036). A threshold decides how many points pass, but says little about where they lie, and any monotone trade-off crosses the other form's default somewhere. The main test is therefore where the gap is smallest, and whether that is where the independently measured κ says.

4.3. Classical baseline

The classical baseline solves the same heat problems with second-order central differences in space on uniform grids whose nodes include every evaluation point, $h = 1/(200m)$ in 1D and $h = 1/(40m)$ in 2D, with $m \in \{1, 2, 4, 8\}$ in 1D and $\{1, 2, 4\}$ in 2D. Time steps are $\Delta t = 0.01/q$ with $q \in \{2, 4, 8, 16\}$, so every evaluation time is hit exactly. Three time schemes are used: plain Crank–Nicolson [25], Crank–Nicolson with Rannacher start-up, in which the first two steps are replaced by four backward Euler half-steps [11], and backward Euler. The reference and the error measure are those of Section 2.1.

5. Results

5.1. Both losses leave the supervised fit

Figure 2 shows the warm-start runs at default weights. With a matching initial condition no arm drifts. With a clashing one both losses drift, at opposite ends of the transient. In 2D the strong form's error is 0.42 at $t = 0.01$ and falls to 0.007 at $t = 1$, while the weak form's is 0.084 early and grows to 0.32 at $t = 1$ (median 0.27 over ten seeds). In 1D the strong form is again worse early (0.28 against 0.083), and the weak form's error is about twice the supervised fit's throughout. Without the initial-condition term neither form learns the transient (error 0.36 to 1.0).

Table 3
Heat 2D, clashing initial condition, warm start: loss terms and error. Means over n seeds; at $t = 1$ also the median and [min–max]. The pointwise PDE residual is 12 times the mean of $(\partial_t u - a\Delta u)^2$ over 16,000 fixed random points. The dual-norm residual was scored on a different set of points.

Model	n	IC loss	PDE residual	$e(0.01)$	$e(1)$: mean, median [min–max]
Supervised start	10	0.253	37.9	0.065	0.040
Weak	10	0.233	52.2	0.084	0.320, 0.269 [0.081–0.721]
Weak, dual norm	10	0.233	53.2	0.084	0.389, 0.330 [0.118–0.795]
Strong	3	0.921	0.046	0.415	0.007, 0.007 [0.006–0.008]

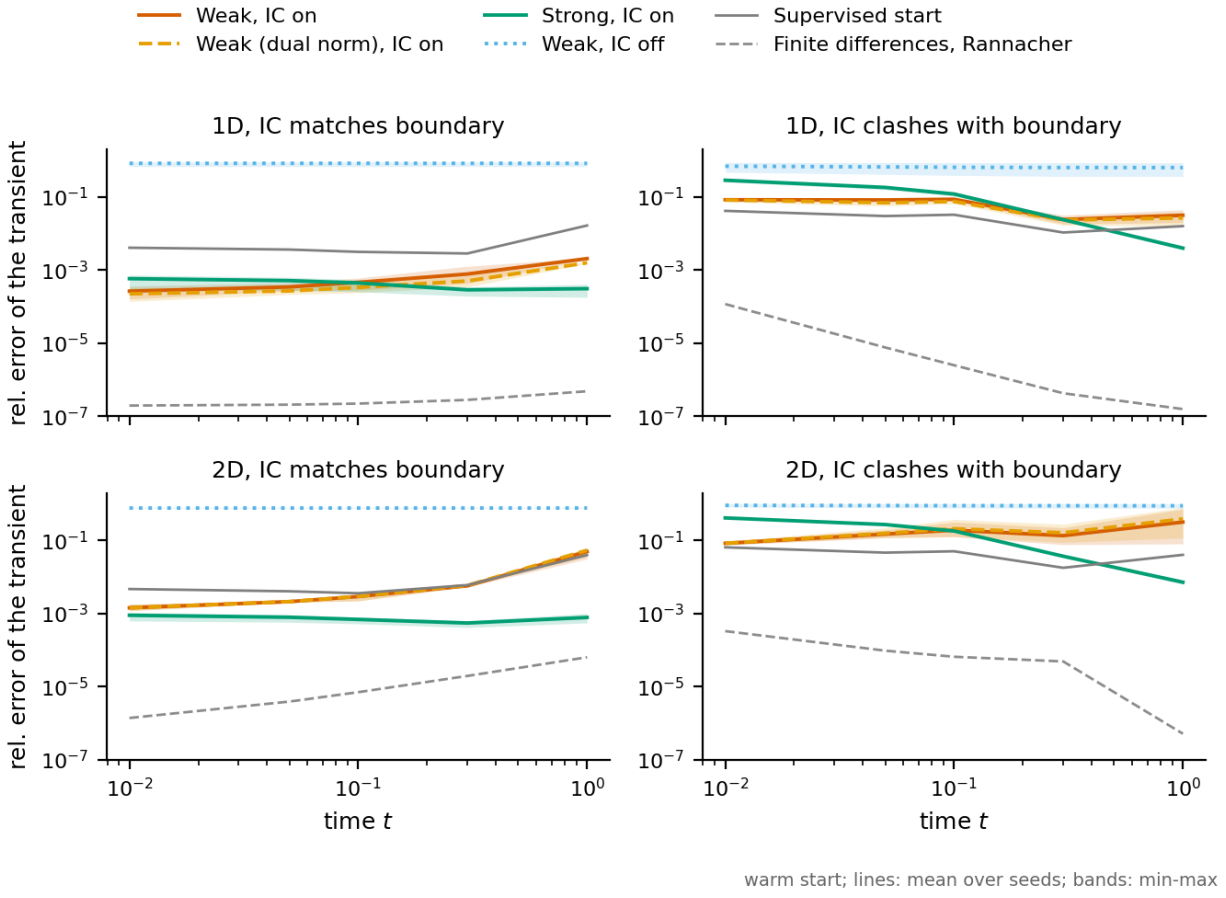


Figure 2: Heat equation, warm start, default weights: relative error of the transient over time. Lines: mean over seeds; bands: min–max. The 2D clashing weak-form arms use ten seeds and all other arms three. Dashed grey: the finest finite-difference solution with Rannacher start-up (Section 5.5). With a matching initial condition (left) no arm drifts. With a clashing one (right) the strong form's error is large early and falls, and the weak form's is small early and grows. The dual-norm weak loss behaves like the plain one. With the initial condition off, the weak form stays far from the transient.

Table 3 shows how each loss resolves the conflict in 2D. The strong form gives up the initial condition (penalty $0.25 \to 0.92$) and repairs the physics (pointwise residual $38 \to 0.046$). The weak form keeps the initial-condition fit and leaves the residual at about 52.

Absolute errors The relative error of Eq. (8) divides by the size of the transient, which shrinks with time. In absolute terms (Table 4) the pattern holds. In 2D the strong form's error peaks at $t = 0.01$ (0.36) and the weak form's at $t = 0.1$ (0.13), and from $t = 0.3$ on the weak form's error is 4 to 44 times the strong form's.

Table 4
Heat, clashing initial condition, warm start, default weights: absolute error, the relative error of Eq. (8) times the RMS of $u - 1$ on the evaluation grid, mean over n seeds. "Transient" is the RMS of $u - 1$ itself.

		$t = 0.01$	$t = 0.05$	$t = 0.1$	$t = 0.3$	$t = 1$
1D	Transient	0.95	0.88	0.82	0.67	0.34
	Strong ($n = 3$)	0.27	0.16	0.098	0.016	0.0013
	Weak ($n = 3$)	0.079	0.072	0.070	0.016	0.010
2D	Transient	0.88	0.76	0.66	0.44	0.11
	Strong ($n = 3$)	0.36	0.21	0.12	0.016	0.00079
	Weak ($n = 10$)	0.074	0.11	0.13	0.060	0.035

The loss prefers the drifted model Over weak-form training in 2D both terms of the objective fall, the initial-condition term from 0.253 to 0.233 and the weak residual from 6.6×10^{-5} to 3.9×10^{-5}, while the error at $t = 1$ rises eightfold, from 0.040 to 0.320. Among the models the network can reach, the loss ranks the drifted one above the supervised fit. A falling training loss is therefore no evidence that the model approaches the solution, and the warm-start test detects what the loss does not.

Seeds and cold starts The late-time 2D weak-form errors vary between seeds (0.081 to 0.72 at $t = 1$ over ten seeds), while the early errors agree closely (standard deviation 0.006 at $t = 0.01$). From random initializations the same split appears: in 2D the strong form is worse at $t = 0.01$ (0.23 against 0.13) and recovers to 0.005, while the weak form reaches 0.59 at $t = 1$.

5.2. The effective weight moves the failure

κ is 2,200 on 1D heat and 2.2×10^5 on 2D heat (Table 2).

A large effective weight is harmless without a clash With a matching initial condition the weak form's initial-condition term has an effective weight of 125 to 637 in 2D, and nothing drifts. The scale gap is as large as under the clash; only the clash is removed. The clash causes the failure, and the effective weight decides where it shows. This separates the effect from loss imbalance as usually studied [12, 13].

Weight sweeps Figure 3 shows the sweeps; the full tables are in the supplementary material. Raising the strong form's weight moves its error from early to late: in 1D its error at $t = 0.01$ falls from 0.28 to 0.078 while its error at $t = 1$ rises from 0.004 to 0.052. Its gap to the weak default is smallest at the measured κ, 0.004 at 2,200 times its default, and under 0.02 from 940 to 3,400 times. Lowering the weak form's weight does the reverse. Its gap to the strong default is smallest at about $2.2/\kappa$ in both 1D (0.016) and 2D (0.014), where the two forms also reach the same initial-condition loss (0.944 against 0.921 in 2D). In 1D the two sweeps trace one curve: a weak run and a strong run whose weights differ by 1,000 to 3,000 agree in their errors and in their loss terms, for example in initial-condition losses of 0.131 and 0.125 and pointwise residuals of 20.1 and 20.0. In 2D the strong form takes on the weak form's behaviour from 10^4 times its default upward, its early error falling from 0.42 to 0.094 and its initial-condition loss reaching the weak form's 0.233. Its closest point, 0.55κ, has a worst gap of 0.096, set by the late-time seed spread of the weak default. A Gaussian-process fit to each sweep's gap places the minimum between about 0.7 and 2.3 times κ in all six sweeps (supplementary material).

Table 5
Weak-form mesh, 1D heat, clashing initial condition, warm start: relative error, mean over seeds 42–44. The strong form is unchanged except for its initial-condition weight, set to κ times its default, with κ measured on each mesh. "Gap" is the largest difference between the two rows over the five times.

E	κ	Arm	$t = 0.01$	$t = 0.05$	$t = 0.1$	$t = 0.3$	$t = 1$	gap
4	490	weak, default	0.057	0.031	0.043	0.021	0.020	0.024
		strong, $\times\kappa$	0.078	0.055	0.056	0.022	0.021	
8	1,300	weak, default	0.065	0.056	0.062	0.023	0.021	0.015
		strong, $\times\kappa$	0.080	0.070	0.073	0.023	0.031	
16	2,200	weak, default	0.083	0.082	0.085	0.024	0.031	0.004
		strong, $\times\kappa$	0.083	0.080	0.083	0.021	0.035	
32	7,100	weak, default	0.087	0.120	0.115	0.032	0.038	0.009
		strong, $\times\kappa$	0.089	0.120	0.113	0.033	0.047	

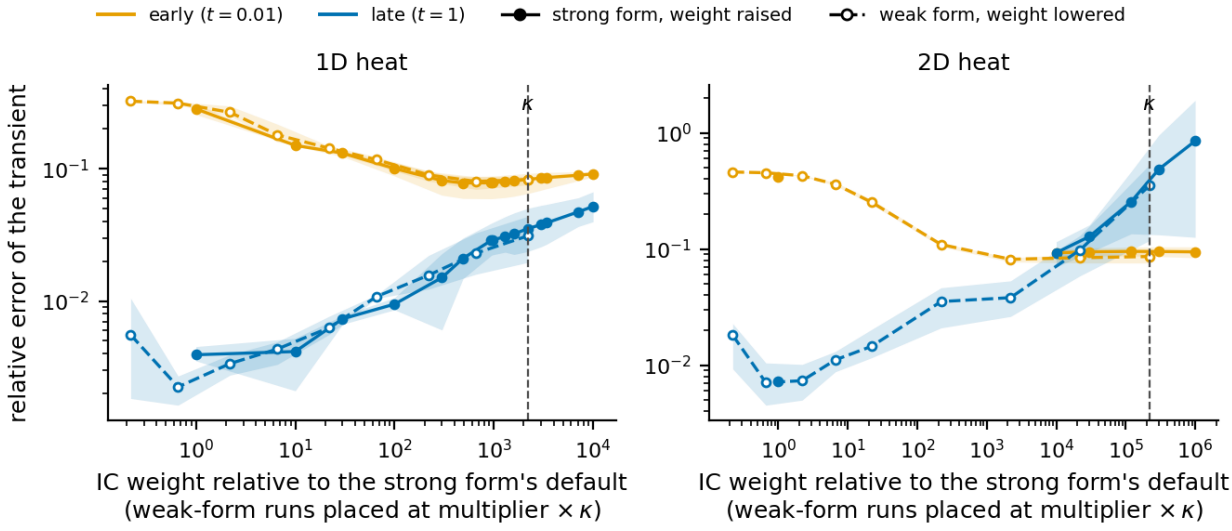


Figure 3: The initial-condition weight moves where each loss fails. Clashing initial condition, warm start, seeds 42–44; left: 1D heat, right: 2D heat. Lines: mean; bands: min–max. Early ($t = 0.01$) and late ($t = 1$) error against the initial-condition weight relative to the strong form's default. Weak-form runs are placed at their multiplier times κ (2,200 in 1D, 2.2×10^5 in 2D). Raising the strong form's weight (filled) moves its error from early to late; lowering the weak form's (open) does the reverse. In 1D the two fall on one curve. Dashed: κ.

5.3. Mesh, order and normalization

Mesh Changing only the weak form's mesh changes κ (Table 2), and the weak form's failure moves with it: its error at $t = 0.05$ rises from 0.031 on 4 elements to 0.120 on 32 (Table 5). The strong form at κ times its default follows it within 0.024 on every mesh. The same loss at the same nominal weight fails differently on a different mesh, as κ predicts.

Order On 16 elements κ varies by a factor of 3.6 across orders $p = 2$ to 10, as Lemma 1 predicts ($11/3 \approx 3.7$), and at each order the weak form at its default weight lands within 0.005 of the strong form at $\kappa(p)$ times its default (Table 6).

Normalization Dividing the weak residual by its element measure multiplies the physics term by E^2, and κ falls by that factor, to $\kappa_{\text{norm}} = 30$, 20, 8.6 and 6.9 on 4 to 32 elements. These values approach the $M = 7$ that Lemma 1 leaves after normalization. Under Adam this is the same as lowering the weak form's initial-condition weight by E^2, and the

Table 6
Test-function order and normalization, 1D heat, clashing initial condition, warm start, seeds 42–44. "Gap" is the largest difference over the five times from the strong form at κ (order) or κ_{norm} (normalized) times its default.

Order p (16 elements)	2	4	6	10
κ	940	1,600	2,200	3,400
gap	0.005	0.003	0.004	0.005
Elements E (normalized)	4	8	16	32
κ_{norm}	30	20	8.6	6.9
gap	0.021	0.024	0.026	0.029

Table 7
Advection–diffusion, clashing initial condition, warm start, mean over seeds 42–44. Top: relative error at default weights. Bottom: gap to the other form's default for each initial-condition multiplier. $\kappa = 2{,}000$.

Default weights	$t = 0.01$	$t = 0.05$	$t = 0.1$	$t = 0.3$	$t = 1$	
Strong	0.171	0.087	0.064	0.050	0.056	
Weak	0.058	0.033	0.048	0.088	0.178	
Strong multiplier	$\times 100$	$\times 300$	$\times 1000$	$\times 2000$	$\times 3000$	$\times 10^4$
Gap to weak default	0.153	0.120	0.066	0.030	0.014	0.070
Weak multiplier	$\times 10^{-4}$	$\times 3{\cdot}10^{-4}$	$\times 5{\cdot}10^{-4}$	$\times 10^{-3}$	$\times 3{\cdot}10^{-3}$	$\times 10^{-2}$
Gap to strong default	0.280	0.119	0.053	0.007	0.044	0.090

failure follows the rescaling. The normalized weak form behaves like the strong form at 7 to 30 times its default, its failure moves from late to early (error at $t = 1$ of 0.005 against 0.031 on 16 elements), and its dependence on the mesh shrinks: the largest spread of its error over the four meshes falls from 0.089 to 0.032. Over all eleven settings of mesh, order and normalization, the weak form matches the strong form at the corresponding κ within 0.03.

5.4. A second equation

The main tests were repeated on the advection–diffusion problem of Eq. (2) with the same trial space, losses and network. With the clash $\kappa = 2{,}000$, against 2,200 for 1D heat on the same mesh, so the ratio belongs to the discretization of the loss rather than to the equation. The defaults split as on heat (Table 7), and the forms meet near the measured ratio. The strong form's gap to the weak default is smallest at 1.5κ (0.014) and the weak form's at $2/\kappa$ (0.007), where the loss terms also agree (initial-condition loss 0.064 against 0.066, pointwise residual 75.2 against 75.0). With a matching initial condition the strong form at κ times its default stays within 0.014 of its start, so a large weight is again harmless when it can be satisfied.

5.5. A classical baseline

Table 8 shows the same clash for a classical discretization. Plain Crank–Nicolson shows the textbook failure under incompatible data: its amplification factor for stiff modes is close to -1, so the error from the jump at $t = 0$ decays slowly [10]. In 1D at $\Delta t = 5 \times 10^{-3}$, refining the mesh raises its error at $t = 0.3$ from 6.6×10^{-4} to 8.1×10^{-3}. Rannacher start-up [11] removes this and restores second order in time, and with matching data plain Crank–Nicolson converges at second order. The finest Rannacher solutions are within 3.3×10^{-4} of the reference at every time.

The comparison isolates the cause. The grid stores the jump at $t = 0$ exactly, with the boundary nodes at 1 and the interior nodes at 0, and Rannacher start-up repairs a time-stepping problem. A smooth trial function pinned to the boundary can approach the jump only through a layer that a fixed network cannot follow, which is why the two loss terms pull against each other. The clash is a property of the trial space and the loss, not of the equation.

5.6. Computational cost

Table 9 gives the cost per training step of each loss and the cost of measuring κ, timed on one GPU with no other jobs running. On the default meshes both losses evaluate the network at the same number of points per step (1,280 in 1D, 16,000 in 2D). Per step, the strong loss needs a forward pass and $n + 1$ backward passes through the network

Table 8
Clashing initial condition: relative error of a finite-difference solver and of the trained models (warm start, default weights). FD rows: second-order central differences; "R" is Crank–Nicolson with Rannacher start-up, "CN" plain Crank–Nicolson. Model rows: mean over n seeds.

	Solver	$t = 0.01$	$t = 0.05$	$t = 0.1$	$t = 0.3$	$t = 1$
1D	R, $h = 1/1600$, $\Delta t = 6.25{\cdot}10^{-4}$	$1.2{\cdot}10^{-4}$	$7.5{\cdot}10^{-6}$	$2.5{\cdot}10^{-6}$	$4.2{\cdot}10^{-7}$	$1.5{\cdot}10^{-7}$
	CN, $h = 1/1600$, $\Delta t = 6.25{\cdot}10^{-4}$	$2.5{\cdot}10^{-3}$	$1.9{\cdot}10^{-4}$	$6.6{\cdot}10^{-5}$	$1.1{\cdot}10^{-6}$	$3.6{\cdot}10^{-8}$
	R, $h = 1/1600$, $\Delta t = 5{\cdot}10^{-3}$	$9.7{\cdot}10^{-3}$	$4.8{\cdot}10^{-4}$	$1.5{\cdot}10^{-4}$	$2.1{\cdot}10^{-5}$	$1.0{\cdot}10^{-5}$
	CN, $h = 1/1600$, $\Delta t = 5{\cdot}10^{-3}$	$5.2{\cdot}10^{-2}$	$1.6{\cdot}10^{-2}$	$5.9{\cdot}10^{-3}$	$8.1{\cdot}10^{-3}$	$9.7{\cdot}10^{-4}$
	Strong ($n = 3$)	0.282	0.180	0.119	0.024	0.004
	Weak ($n = 3$)	0.083	0.082	0.085	0.024	0.031
2D	R, $h = 1/160$, $\Delta t = 6.25{\cdot}10^{-4}$	$3.3{\cdot}10^{-4}$	$9.7{\cdot}10^{-5}$	$6.6{\cdot}10^{-5}$	$4.9{\cdot}10^{-5}$	$5.2{\cdot}10^{-7}$
	Strong ($n = 3$)	0.415	0.273	0.182	0.037	0.007
	Weak ($n = 10$)	0.084	0.150	0.192	0.136	0.320

Table 9
Computational cost on one NVIDIA A100-SXM4-80GB, double precision, no other jobs running. Per-step time: mean over 300 steps after 30 warm-up steps, physics and initial-condition terms with backward pass and Adam update. Run times are the per-step time multiplied by the steps of a warm (10,000) or cold (15,000) run. κ: one measurement with $D = 100$ gradient draws of each loss term at one seed. Finite differences: finest Rannacher solve on a CPU (Section 5.5).

	Loss	points/step	ms/step	peak MB	warm run (min)	cold run (min)
1D	strong	1,280	19.1	128	3.19	4.79
1D	weak	1,280	12.3	93.2	2.04	3.07
1D	weak, dual norm	1,280	13.2	93.2	2.2	3.29
2D	strong	16,000	30.8	1.27e+03	5.13	7.7
2D	weak	16,000	14.4	396	2.39	3.59
2D	weak, dual norm	16,000	14.7	396	2.46	3.69
1D	κ measurement: 2.68 s					
2D	κ measurement: 3.93 s					

at its $|Q|$ points, one for the first derivatives and one per dimension for the second. The weak loss needs a forward pass and one backward pass at its $10E^nQ^n$ quadrature points, with Q points per dimension, followed by a contraction with the test functions that costs $O(10E^nMQ^n)$ operations and is small next to the network passes. The weak loss is therefore cheaper per step, by a factor of 1.6 in 1D and 2.1 in 2D, and in 2D it uses about a third of the memory. A warm-started run takes 2 to 5 minutes. Measuring κ takes $D = 100$ gradient evaluations of each loss term at a single point in parameter space, 3 to 4 s per seed, about 2% of a warm weak-form run. On fine meshes Lemma 1 gives κ with no measurement at all.

The finite-difference solver of Section 5.5 takes 0.16 s in 1D and about 8 s in 2D on a CPU. Physics-informed models are used where such a solver is not a substitute, and the warm-start test of Section 5.1 can use a classical solve of this kind as its reference.

6. Remedies

Intermediate weights Since the weight moves the failure between the two ends of the transient, intermediate weights reduce the worst error. The strong form at 100 to 300 times its default keeps its 1D error at or below 0.10 at every time, against 0.28 at the default, and the weak form at 10^{-3} times its default in 2D stays at or below 0.11, against 0.351 at $t = 1$ at the default (three-seed means, as in the sweep). The trade-off itself remains, as Proposition 1(b) predicts for the loss terms.

Learning-rate annealing Learning-rate annealing [12] resets the initial-condition weight during training from the ratio of the largest physics gradient to the mean initial-condition gradient (Table 10). It sets large weights in every case, 2×10^4 to 1.3×10^5 times the default for the 1D strong form without the clash and 2×10^7 to 5×10^7 times with it.

Table 10
Learning-rate annealing of the initial-condition weight [12], warm start: relative error, mean over seeds 42–44. "Change" is the largest difference from the same form at its default weight over the five times.

	IC	Form	$t = 0.01$	$t = 0.05$	$t = 0.1$	$t = 0.3$	$t = 1$	change
1D	matching	weak	0.001	0.002	0.003	0.006	0.004	0.005
		strong	0.002	0.002	0.003	0.005	0.003	0.005
	clashing	weak	0.084	0.247	0.458	0.807	0.743	0.783
		strong	0.097	0.380	0.671	0.977	0.447	0.953
2D	matching	weak	0.003	0.004	0.004	0.007	0.037	0.013
		strong	0.003	0.003	0.004	0.005	0.018	0.017
	clashing	weak	0.087	0.107	0.102	0.048	0.179	0.172
		strong	0.095	0.166	0.198	0.081	0.451	0.444

Without the clash this does no harm, and every warm-started arm stays within 0.017 of its default. With the clash it moves the failure rather than removing it: the annealed 2D strong form lands next to its sweep point at 1.2×10^5 times its default, and in 1D both forms are pushed beyond the sweeps, to errors near 1 mid-transient.

Ramping the boundary Switching the boundary on smoothly, $g(t) = 1 - e^{-t/\tau}$, removes the jump at $t = 0$. On 1D heat with $\tau = 0.05$ and 0.01, the weak form and the strong form at 630 and 870 times its default (a third to a half of $\kappa = 1{,}900$) stay within 0.05 of their supervised starts at every time, while the strong form at its default weight still gives up the initial condition (changes of 0.095 and 0.146). A quick ramp leaves a steep layer at the corner, so the weight still decides the outcome.

Dual-norm weighting The dual-norm loss [17] on the same test space does not remove the drift. In 2D it is slightly worse than the plain weak loss (0.389 against 0.320 at $t = 1$, worse in nine of ten paired seeds, paired t-test $p = 0.0007$). Its guarantee needs a richer test space than the one used here [17].

Masking the initial condition Dropping initial-condition points within a band of the boundary [16] reduces the 2D weak form's late error ($0.351 \to 0.107$ warm, $0.485 \to 0.262$ cold) but raises the strong form's ($0.007 \to 0.066$ warm). Removing the initial condition stops the drift but loses the transient.

Normalization Normalizing the weak residual by element measure is the remedy for carrying weights between loss forms and meshes (Section 5.3).

7. Discussion

What the effective weight does When the two terms cannot both be made small, the initial-condition term keeps pulling, and the effective weight decides which gives way. When it is small, the model gives up the initial condition and repairs the physics, so it is wrong early and recovers. When it is large, the model holds the initial condition and leaves the physics unrepaired, so it drifts late. Proposition 1 shows this trade-off for the loss terms of a linear trial space, and the sweeps show it for the network: the switch is gradual, and annealing, which raises the weight automatically, moves along it.

Weights do not transfer The same nominal weight means different things in a weak and a strong loss, and in a weak loss on different meshes. On fine meshes the factor is ME^{2n} (Lemma 1) and can be computed before training; on coarser meshes $(h/\ell)^2$ at a fit shows how far it moves. Normalizing the weak residual by element measure removes the E^{2n} part. Under a clash this matters more than usual, because the effective weight decides which part of the solution is lost.

A test that can be run Fit the network to a reference, train it on the physics loss alone, and see whether its error grows. On these problems the test exposes the failure in both forms while the training loss falls; on 2D heat the weak

objective falls while the late error grows eightfold. A classical solve, such as the finite-difference solver of Section 5.5, can serve as the reference.

Higher dimensions The factor grows quickly with dimension. In 3D with 8 elements per dimension and $p = 4$, the formula of Lemma 1 gives $M E^{2n} = 125 \cdot 8^6 \approx 3.3 \times 10^7$. Normalization by element measure removes the E^{2n} part at no cost.

Where the clash arises Any model that builds a Dirichlet boundary into the trial space and keeps the initial condition soft meets the clash when the boundary is switched on suddenly: a wall brought to a new temperature, a step load, a surface exposed to a new environment at $t = 0$. The cheap actions are to smooth the switch-on where the problem allows, to normalize the weak residual by element measure, to compute or measure κ before reusing weights, and to run the warm-start test when a reference exists.

Scope The model problems are linear, on the unit square, with exact solutions, so every claim is checked against the exact answer. One network architecture and one family of test functions are used, the loss is weak in space and collocated in time, and the mesh, order and normalization interventions are 1D. Training uses Adam. Most arms use three seeds and the 2D weak-form defaults ten.

8. Conclusion

When a hard-coded boundary contradicts a soft initial condition, a network of fixed size cannot make the initial-condition penalty small without hurting the physics, and its pull moves both weak-form and strong-form training away from a supervised fit to the exact solution, at opposite ends of the transient. Where the model goes wrong is set by the effective weight of the penalty. The weak-form discretization multiplies that weight by $\kappa = M E^{2n}$ on fine meshes, with a correction that tracks the variation of the residual across an element. Weight sweeps and changes of mesh and test-function order move each form's failure as κ predicts, on heat and on advection–diffusion. A finite-difference solver with Rannacher start-up resolves the same clash, which places the problem in the trial space and the loss. In practice: smooth the switch-on where the problem allows, normalize the weak residual by element measure, compute or measure κ before carrying weights between loss forms or meshes, and test a physics loss by starting from a supervised fit and checking whether its error grows.

CRediT authorship contribution statement

Ayush Gouda: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. **Swaraag Hebbar N:** Writing – review & editing. **Aditya Prakash:** Writing – review & editing. **Sudheendra B S:** Supervision, Writing – review & editing.

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Data availability

The code and all logged results, including per-seed values for every number in the text, are available at `https://github.com/AyushG-1210/Pipeline-Digital-Twin`.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used an AI assistant to help edit the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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